GARRETT TURBINE ENGINE COMPANY
PHOENIX, ARIZONA

TO: HOLDERS OF TURBOFAN ENGINE TROUBLE SHOOTING TESTER ASSEMBLY, GROUND EQUIPMENT MANUAL REPORT NO. 72-GE-01, REVISED MARCH 5, 1980.

THIS REVISED PUBLICATION IS ISSUED FOR USE IN SUPPORT OF THE FOLLOWING AIRCRAFT.

TESTER ASSY PART NO. 289900 -1	-2	-3	ENGINE MODEL NO.	AIRCRAFT APPLICATION
X	X	X	TFE731-3-1E	731 Jetstar
X	X	X	TFE731-3-1G	Westwind 1124
X	X	X	TFE731-3-1F	Jetstar II
	X	X	TFE731-2-1C	Falcon 10
	X	X	TFE731-2-2B	Learjet 35/36 Series
	X	X	TFE731-2-2J	Casa 101
	X	X	TFE731-3-1C	Dassault Falcon 50
	X	X	TFE731-3-1D	Sabreliner 65/65A
	X	X	TFE731-3-1H	British Aerospace HS125-SER
	X	X	TFE731-3A-2B	Learjet
	X	X	TFE731-3A-2B1	Learjet
		X	TFE731-3B-100S	Cessna Citation III
		X	TFE731-3R-1D	Sabreliner 65/65A
		X	TFE731-3R-1H	British Aerospace HS125-SER

REVISION NO. 3 DATED JULY 13, 1982

This is a COMPLETE revision. Due to the extent of the changes involved in this revision, this publication has been reprinted in its entirety. However, pages or prior issues, which are not affected, retain previous revision dates. Please remove and discard all pages of prior issues and replace with Revision 3, dated July 13, 1982.

THIS DOCUMENT IS COMPLETE
RETAIN TABS

HIGHLIGHTS

Subject/Pages	Description of Changes

NOTE: When the letter R appears in the left hand margin opposite the page number, it indicates that a change of a non-technical nature (i.e. typographical error, relocated material, etc.) has been made to that page. These pages are not listed in the Highlights.

72-GE-01

Subject/Pages	Description of Changes
Title Page	Changed revision date. Deleted Engine Part No. Column.
Service Bulletin List	Added Revision No. 3 to Service Bulletin No. 2.
Chapter 1 General	Revised Figure 1 and 2 to include guard for switch S-12.
Chapter 2 Maintenance	Revised Figures 1, 2 and 3 to correct drawing error.
Chapter 4 Illustrated Parts List	Revised as required to update to current configuration.
	Revised Figures 1 and 3 to illustrate guard for switch S12. Figure 2 to illustrate added cable.

GARRETT TURBINE ENGINE COMPANY
PHOENIX, ARIZONA

TURBOFAN ENGINE TROUBLE SHOOTING TESTER ASSEMBLY

PART NO. 289900-1
289900-2
289900-3

GROUND EQUIPMENT MANUAL WITH ILLUSTRATED PARTS LIST

REPORT NO. 72-GE-01
May 15/73
Revised July 13/82

LIST OF EFFECTIVE PAGES
72-GE-01

SUBJECT	PAGE	DATE
Title Page	-	July 13/82
List of Effective Pages	1	July 13/82
	2	July 13/82
Record of Revisions	-	July 13/82
Record of Temporary Revisions	-	Mar 5/80
Service Bulletin List	-	July 13/82
Introduction	1	July 13/82
	2	Blank
Table of Contents	-	Mar 5/80
1-0 Contents	1	Mar 5/80
	2	Blank
1-1	1	July 13/82
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	4	Blank
1-2	1	July 13/82
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1-3	1	Mar 5/80
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	9 FO	July 13/82
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Numerical Index		
Alpha	3	July 13/82
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Detailed Parts List		
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	2	July 13/82
	3	July 13/82

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289900

RECORD OF REVISIONS

	REV NO.	ISSUE DATE	DATE INSERTED	BY	REV NO.	ISSUE DATE	DATE INSERTED	BY
	1	Jan 21/77						
	2	Mar 5/80						
R	3	July 13/82						

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RECORD OF TEMPORARY REVISIONS

REV NO.	CHAP/SEC/ SUBJ PAGE NO.	DATE ISSUED	DATE INSERTED	BY	DATE REMOVED	BY	INCORPO-RATED INTO MANUAL BY REV NO.

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SERVICE BULLETIN LIST

	Service Bulletin	Incorp.	Title
	289900 SB No. 1, Revision 1	Mar 5/80	TFE TESTER ASSEMBLY - Modify TFE Trouble Shooting Tester Assembly to be compatible with Fuel Control Part No. 3070800-7.
	289900 SB No. 2, Revision 2	Mar 5/80	TFE TESTER ASSEMBLY - Modify Turbofan Engine Tester Assembly to be compatible with Electronic Computer Incorporating APR.
R R R R R	289900 SB No: 2, Revision 3	July 13/82	TFE TESTER ASSEMBly - Provide expanded rework procedures for Turbofan Engine Tester Assembly to be compatable with Electronic Computer Incorporating APR.

INTRODUCTION

This publication provides instructions for the equipment listed in Table 1, Equipment Identification List for Turbofan Engine Trouble Shooting Tester, manufactured by The Garrett Turbine Engine Company, A Division of The Garrett Corporation, P.O. Box 29003 Phoenix, AZ 85038.

Table 1. Equipment Identification List for Turbofan Engine Trouble Shooting Tester.

Code Symbol*	Part No.	Service Bulletin
A	289900-1	
B	289900-2	289900 No. 1
C	289900-3	289900 No. 2

*These alpha code symbols are used in the Effectivity Code Column of the Illustrated Parts List, and in the text, to show differences between equipment configurations.

This publication is prepared in accordance with Air Transport Association Specification No. 101. Definitions of words and terms used in the publication are as stipulated therein.

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GROUND EQUIPMENT MANUAL
289900

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CHAPTER 1

TABLE OF CONTENTS

GENERAL INFORMATION AND OPERATING INSTRUCTIONS

1. Description

A. Purpose of Manual

This manual contains general information; operating, maintenance, and repair instructions; and an illustrated parts list for the turbofan engine trouble shooting tester manufactured by Garrett Turbine Engine Company, A Division of the Garrett Corporation, P.O. Box 29003, Phoenix, AZ 85038.

B. Functional Description of the Turbofan Engine Trouble Shooting Tester

The turbofan engine trouble shooting tester (see 1-1, figure 1) hereafter referred to as the tester, is used in conjunction with an applicable engine maintenance or overhaul manual to provide flight line fault isolation of a line replaceable unit when trouble shooting turbofan engines. Two cables, approximately 60 inches in length, are included in the tester case. One cable connects between the tester and the engine electronic computer connector. The other cable connects the tester to the disconnected end of the engine computer wiring harness. Switches, indicators, and test jacks are used to isolate specific circuits and provide points for testing voltage or resistance of the wiring or components. A 3.0 volt dc external battery source is required to supply temperature simulation when testing interstage turbine temperature (ITT).

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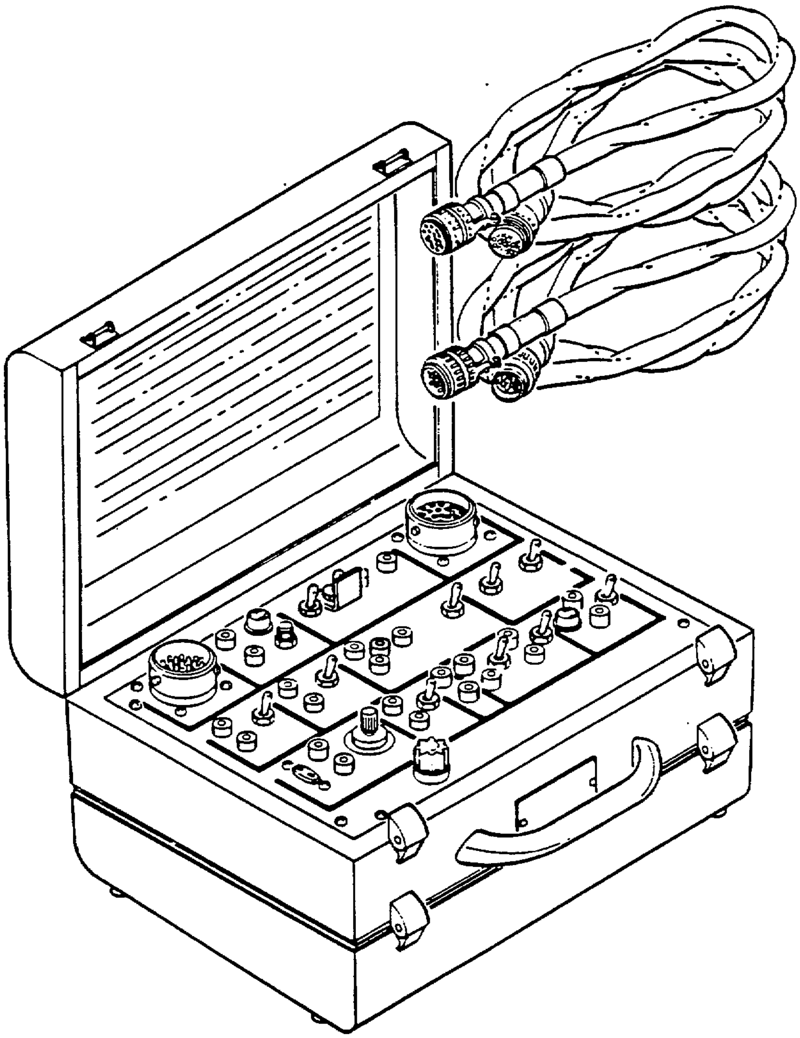


Turbofan Engine Trouble Shooting Tester Assembly
Figure 1

1. C. Physical Description of Major Components. (See 1-1, Figure 1)

(1) General

Major components of the tester include a case assembly, panel assembly and two connecting cable assemblies. Not included, but required, are an external 28 volt dc power source and an external 3.0 volt dc battery. Switches, jacks, plugs, indicating lamps, and controls are clearly marked to indicate circuits.

(2) Case Assembly

The case is made of metal and has two removable hinged lids. One lid covers the panel assembly and the other provides a compartment to store the two cable assemblies. Rubber bumpers on the bottom of the case are used to cushion and insulate the tester when in use. An identification marker, reflecting the electrical schematic of the tester, is located in the upper lid.

(3) Panel Assembly

The visible side of the panel assembly contains and identifies the switches, test jacks, indicating lights, controls and two electrical connectors. The underside of the panel contains an electronic terminal board, resistors, diodes, wires, splices, and a switching transistor.

(4) Cable Assemblies

Two cable assemblies are required to connect the tester into the engine computer electrical circuit. One of the cables is connected between the connector on the computer and the connector marked COMPUTER on the tester panel. The other cable is attached between the removed computer wiring harness connector and the connector marker ENGINE on the tester panel. Chromel pins or sockets are found at the No. 33 location in all connectors. Alumel pins or sockets are used at the No. 34 location in all connectors. Unused connector wire locations shall have pins and plugs installed.

(5) External Power Source

The tester utilizes a 28 volt dc power source supplied by the aircraft electrical system. A 3.0 volt dc external power source (battery) is plugged into the jacks in the Interstage Turbine Temperature (ITT) section for testing ITT.

2. Operation (See 1-2, Figure 1)

A. General

The description of the controls and their functions is intended for familiarization only. Refer to applicable engine maintenance or overhaul manual for tester operation with a specific engine.

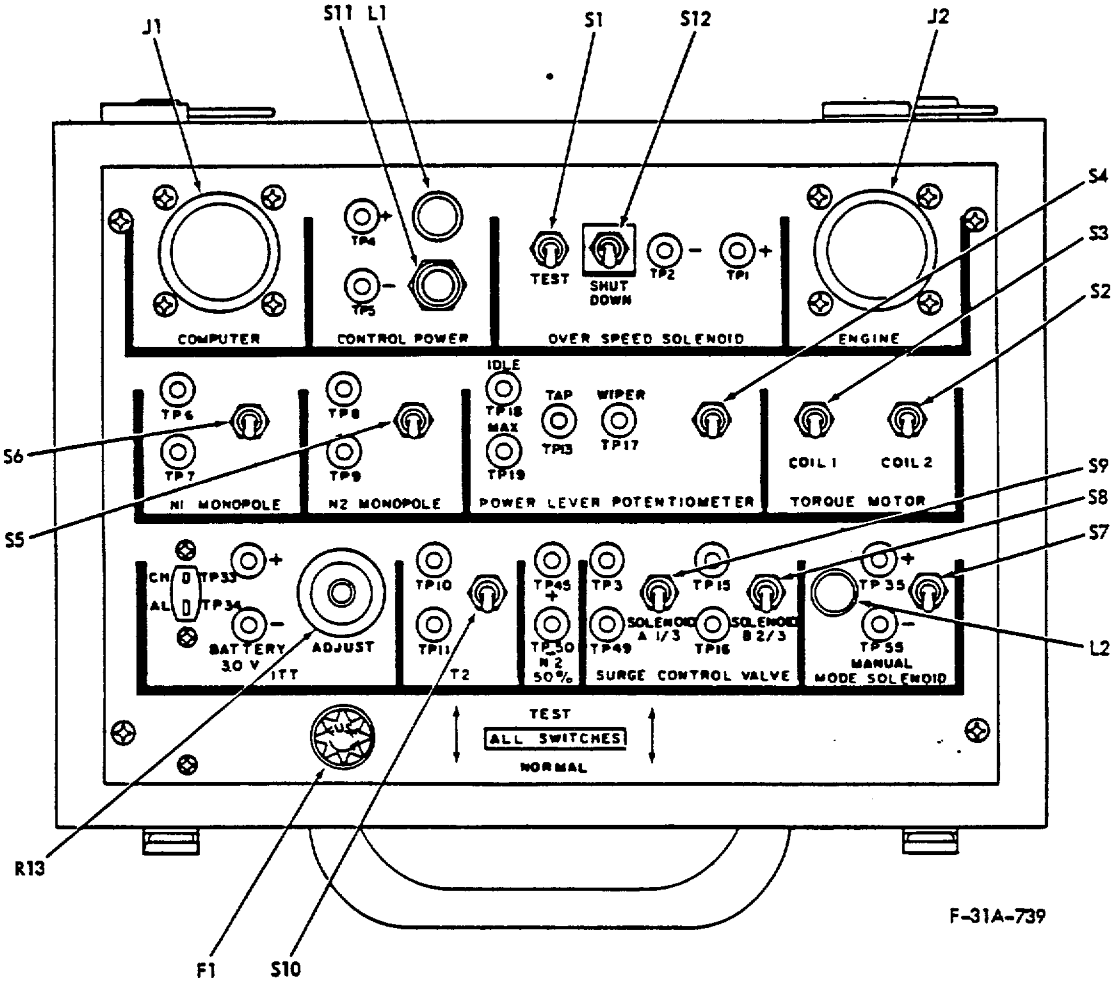
R CAUTION: DO NOT CONFUSE SWITCH TITLE DECALS WITH SWITCH
R POSITIONS. ALL SWITCHES ARE IN NORMAL POSITION WHEN
R POINTING TOWARD BOTTOM OF PANEL AND IN THE TEST
R POSITION WHEN POINTING TOWARD THE TOP OF THE PANEL.

B. Control and Indicator Functions

Refer to Table 1 for familiarization of control and indicator functions.

Table 1. Control/Indicator Functions

Control/Indicator	Equipment Designator	Function
Fuse (10 Amp)	F1	Provides overload protection for tester circuits.
Connector	J1	Connector for test cable between tester and electronic computer
Connector	J2	Connects test cable between tester and computer wiring harness.
Indicator	L1	Indicates that power is being supplied to the engine electronic computer.
Switch	S11	Push button switch supplies power to engine electronic computer when test switches S1 through S10 are set to the TEST position. Switch remains set only when pressed.



R

Turbofan Engine Tester Controls and Indicators (Typical)
Figure 1

Table 1. Control/Indicator Functions (Cont)

Control/Indicator	Equipment Designator	Function
Jacks	TP4-TP5	Provide test points for measuring voltage to engine electronic computer.
Switch	S1	When set to TEST, disconnects engine overspeed solenoid and connects tester simulated load to circuit.
(Code C) Switch	S12	When set to TEST, applies 28 volt dc power to overspeed solenoid in engine fuel control, shutting off fuel to engine.
Jacks	TP1, TP2	Provide test points for measuring resistance of, or voltage to, engine overspeed solenoid.
Jacks	TP6, TP7	Provides test points for measuring resistance of, or voltage from, engine N1 monopole transducer.
Switch	S6	When set to TEST, disconnects engine N1 transducer and connects tester simulated load to circuit.
Jacks	TP8, TP9	Provide test points for measuring resistance of, or voltage from, engine N2 monopole transducer.
Switch	S5	When set to TEST, disconnects engine N2 transducer and connects tester simulated load to circuit.

Table 1. Control/Indicator Functions (Cont)

Control/Indicator	Equipment Designator	Function
(Code A) Jacks	TP17, TP18, TP19	Provide test points for measuring resistance of, or voltage to, engine power lever potentiometer.
(Codes B & C) Jacks	TP13, TP17, TP18, TP19	Provide test points for measuring resistance of, or voltage to, engine power lever potentiometer.
Switch	S4	When set to TEST, disconnects engine power lever potentiometer and connects tester simulated load to circuit.
Switch	S2	When set to TEST, disconnects engine torque motor coil No. two and connects tester simulated load to circuit.
Switch	S3	When set to TEST, disconnects engine torque motor coil No. one and connects tester cimulated load to circuit.
Plug	TP33, TP34	Provide test points for measuring resistance of engine interstage turbine temperature (ITT) and a connection for a temperature indicator to check operation of the ITT.
Jacks 3.0 volt dc External Battery (+, -)	TP+, TP-	Provide for connection of external 3.0 volt dc battery which supplies power for adjustment of ITT signal supplied to engine electronic computer.

Table 1. Control/Indicator Functions (Cont)

Control/Indicator	Equipment Designator	Function
Variable Resistor	R13	Varies simulated ITT signal to engine electronic computer.
Jacks	TP10, TP11	Provide test points for measuring resistance of, or voltage to, engine T2 sensor.
Switch	S10	When set to TEST, disconnects engine T2 sensor and connects tester simulated load to circuit.
Jacks	TP45, TP50	Provide test points for measuring resistance of, or voltage from, N2 50% speed switch coil.
Jacks	TP3, TP49	Provide test points for measuring resistance of, or voltage to, engine surge control valve solenoid A.
Switch	S9	When set to TEST, disconnects engine surge control valve A and connects tester simulated load to circuit.
Jacks	TP15, TP16	Provide test points for measuring resistance of, or voltage to, engine surge control valve solenoid B.
Switch	S8	When set to TEST, disconnects engine surge control valve solenoid B and connects tester simulated load to circuit.

Table 1. Control/Indicator Functions (Cont)

Control/Indicator	Equipment Designator	Function
Jacks	TP35, TP55	Provides test points for measuring the resistance of, or voltage to, the engine manual mode solenoid.
Switch	S7	When set to TEST, disconnects engine manual mode solenoid and connects tester simulated load to circuit.

2. C. Perform Preliminary Test Procedures

R CAUTION: ENSURE THAT AIRCRAFT ELECTRIC POWER IS NOT APPLIED
R TO ENGINE ELECTRONIC COMPUTER BEFORE DISCONNECTING
R WIRING HARNESS TO COMPUTER. ALSO, ENSURE THAT VARI-
R ABLE RESISTOR (R13) IN THE ITT SECTION IS IN A FULL
R COUNTER-CLOCKWISE (ZERO) POSITION.

(1) Place tester within five feet of electronic computer.

(2) Remove cables from lower lid. Close lower lid and set tester down on rubber bumpers.

(3) Remove lid covering panel assembly.

(4) Disconnect electronic computer wiring harness from J1 connector on engine electronic computer.

(5) Connect tester electrical cable with socket connectors between J1 connector on panel assembly and J1 connector on electronic computer.

(6) Connect other tester electrical cable with pin connectors between J2 connector on panel assembly and P11 connector on disconnected engine computer wiring harness.

(7) When engine testing requires a simulated ITT signal, set variable resistor (R13) to zero, then connect an external 3.0 vdc battery to the (+) and (-) jacks in the ITT section.

D. Perform Test Procedures

Consult applicable engine manual for conducting fault isolation tests on specific engines.

GENERAL INFORMATION AND OPERATING INSTRUCTIONS

3. Specifications and Capabilities

A. Leading particulars of the tester are listed in Table 1.

Table 1. Leading Particulars

Type of Equipment	Portable Engine Tester
Operating Mode	Engine Component Simulation
Operating Power Supply (External Source)	28 volt dc
Battery Power, ITT Section (External Source)	3.0 volt dc
Dimensions (approximately)	
Length	12 inches
Width	9 inches
Height	6 inches
Cable Length	5 feet

B. Capabilities (See 1-3, Figure 1)

(1) General.

The tester assembly provides simulated loads to check the following engine components: overspeed solenoid, N1 monopole (transducer), N2 monopole (transducer), power lever potentiometer, manual mode solenoid, torque motor, T2 sensor (Inlet Air Temperature) and surge control valve.

A circuit is included to supply a simulated ITT signal in conjunction with the engine thermocouple harness. It also monitors engine electronic computer operating mode and controls power to the engine electronic computer during the test.

(2) Specific Circuits.

(a) Overspeed Solenoid Circuit - Switch S1, in the NORMAL position, connects engine overspeed solenoid for normal operation. In TEST position, resistor R2 provides simulated load. Test points TP1 and TP2 allow checking engine overspeed solenoid resistance or applied voltage.

3. B. (2) (b) (Code C) Switch S12 in TEST position connects TP1 with TP4 to provide 28 vdc power to engine overspeed solenoid, shutting off fuel flow from fuel control.

(c) N1 Monopole (Transducer) Circuit - Switch S6 in NORMAL position, connects engine N1 transducer for normal operation. In TEST position, resistor R6 provides simulated load. Test points TP6 and TP7 allow checking engine N1 transducer output signal.

(d) N2 Monopole (Transducer) Circuit - Switch S5 in NORMAL position, connects engine N2 transducer for normal operation. In TEST position, resistor R5 provides simulated load. Test points TP8 and TP9 allow checking engine N2 transducer resistance or output signal.

(e) Power Lever Potentiometer Circuit - Switch S4, in NORMAL position, connects engine power lever potentiometer for normal operation. In TEST position, resistors R7 and R8 provide simulated load. Resistor R11 in circuit to TP17 protects power lever potentiometer from damage during testing.

(Code A) Test points TP17, TP18 and TP19 allow checking engine power lever resistance or applied voltage.

(Codes B, C) Test points TP13, TP17, TP18 and TP19 allow checking engine power lever resistance or applied voltage.

(f) Manual Mode Solenoid Circuit - Switch S7 in NORMAL position connects engine manual mode solenoid for normal operation. In the TEST position, resistor R4 provides simulated load. Test points TP35 and TP55 allow checking manual mode solenoid or applied voltage. Indicator L2 illuminates whenever engine electronic computer is in normal (automatic) mode.

(g) Torque Motor Circuit - Switch S2 in NORMAL position connects engine torque motor coil No. two for normal operation. In TEST position, resistor R10 provides simulated load. Switch S3 in NORMAL position connects engine torque motor coil No. one for normal operation. In TEST position, resistor R9 provides simulated load.

(h) T2 Sensor Circuit - Switch S10 in NORMAL position connects engine inlet temperature sensor for normal operation. In TEST position, resistor R12 provides simulated load. Test points TP10 and TP11 allow checking engine inlet temperature resistance or applied voltage.

3. B. (2) (i) Surge Control Valve Circuit - Switch S8 in NORMAL position connects engine control valve solenoid B for normal operation. In TEST position, resistor R1 provides simulated load. Switch S9 in NORMAL position connects engine surge control valve solenoid A for normal operation. In TEST position, resistor R3 provides simulated load.

(j) ITT Circuit - Transistor Q1, variable resistor R13, associated components and externally connected 3.0 volt dc battery allow adjustment of ITT signal supplied to engine electronic computer. Test points TP33 and TP34 are used with a temperature indicating device during operational tests.

(k) Control Power Circuit - Electrical power for engine electronic computer is interlocked through the NORMAL position contacts of switches S1 through S10 to allow normal engine operation with tester installed. Whenever this path is interrupted by setting any of switches S1 through S10 to TEST position, switch S11 must be depressed to supply electric power to the electronic computer. Indicator L1 illuminates whenever electrical power from the engine is available to the electronic computer. Test points TP4 and TP5 provide points to check engine voltage supplied to electronic computer.

GENERAL INFORMATION AND OPERATING INSTRUCTIONS

R 4. Shipping

R A. Preparation

R (1) Place cable assemblies in lower lid of case using care not to
R damage electric wires or connectors. Close lid and fasten latches.

R (2) Ensure all switches are in NORMAL position. Close upper lid and
R fasten latches.

R CAUTION: DO NOT PLACE EXTERNAL 3.0 VDC BATTERY IN TESTER
R CASE FOR SHIPMENT. ACID DAMAGE MAY OCCUR.

R B. Packaging

R Place tester assembly in suitable container for shipping.

5. Storage

A. Cables

After use, place cable assemblies in lower lid of case using care not to damage electric wires or connectors. Close lid and fasten latches.

B. Switches

Ensure all switches are in NORMAL position. Close upper lid and fasten latches.

CAUTION: DO NOT STORE EXTERNAL 3.0 VOLT DC BATTERY IN TESTER CASE AS ACID DAMAGE MAY OCCUR.

C. Location

Store tester in an area that affords protection from extreme temperatures, high humidity, corrosive fumes or dust, and physical damage.

CHAPTER 2

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MAINTENANCE

1. Servicing

A. Observe General Servicing Instructions

The Turbofan Engine Trouble Shooting Tester does not require routine servicing. However, cleaning tester periodically in accordance with the following procedures may extend tester life and accuracy.

B. Perform Cleaning Procedure

Table 1. Cleaning Materials and Compounds

Material or Compound	Manufacturer
NOTE: Equivalent substitutes may be used for listed item.	
Abrasive Cloth (Federal Specification P-C-451)	Commercially available
R Solvent (Federal Specification P-D-680, Type I)	Commercially available
Solvent, trichlorotrifluoroethane (MS-180 Freon)	Miller-Stephenson Chemical Co. Inc. P.O. Box 628, 16 Sugar Hollow Rd, Danburry, CT 06810

(1) Clean tester case.

(a) Remove moisture, condensation, or dust with a clean, dry, lint-free cloth.

WARNING: USE SOLVENT IN A WELL VENTILATED AREA. KEEP AWAY FROM FLAME. AVOID BREATHING VAPORS.

(b) Remove oil or grease with a lint-free cloth, dampened with solvent (Federal Specification, P-D-680, Type I).

(c) Remove scratches with abrasive cloth (Federal Specification P-C-451).

NOTE: Painting of the tester is not required.

1. B. (2) Clean electrical connectors.

(a) Clean electrical connectors using trichlorotrifluoroethane (MS-180 Freon).

(b) Wipe leads clean with lint-free cloth.

MAINTENANCE

2. Trouble Shooting

A. Observe General Trouble Shooting Instructions

(1) Trouble shooting procedures are based on suspected malfunctions of tester or cable circuits and are presented in a symptom - cause - remedy format, except for the ITT circuit, which has a component simulation procedure. Equipment listed in Table 1 may be used to perform trouble shooting procedures.

(2) When a malfunction occurs in any section except the ITT circuit of tester, refer to Table 2, symptom column, to determine the components involved. Analyze the probable cause(s) and check until a remedy is determined. Perform the required remedy, then repeat test for condition encountered. Trouble shooting of the ITT circuit is covered in paragraph C.

Table 1. Equipment and Materials

Equipment	Manufacturer/Description
NOTE: Equivalent substitutes may be used for listed item.	
Millivoltmeter (Model 387)	Simpson Electric Company, 5200 W. Kinzie Street, Chicago, IL, 60644 Used to measure voltage of ITT signal simulation circuit.
Multimeter (Type 3-3022)	Triplett Corporation, 286 Harmon Place, Bluffton, OH 45817 Measures resistance of electrical components.
Battery (3.0 volts dc)	Commercially available Provides power required to perform ITT component simulation test when connected to jacks TP33 (+) and TP34 (-).

R CAUTION: DO NOT CONFUSE SWITCH TITLE DECALS WITH SWITCH
R POSITIONS. ALL SWITCHES ARE IN NORMAL POSITION WHEN
R POINTING TOWARD BOTTOM OF PANEL AND IN TEST POSITION
R WHEN POINTING TOWARD TOP OF PANEL.

2. B. Perform Turbofan Engine Trouble Shooting Tester Assembly Fault Isolation Tests

Refer to Table 2 for fault isolation information to aid in trouble shooting.

Table 2. Fault Isolation Information

Symptom	Probable Cause	Remedy
NOTE: Refer to 2-2, Figures 1-3 for aid in troubleshooting.		
Excessive resistance J2-4 to J1-4 with switches S1-S10 NORMAL and S11 not depressed.	Defective fuse (F1). Defective switch or switches (S1-S10).	Replace fuse. Replace defective switch or switches.
Excessive resistance J2-4 to J1-4 with CONTROL POWER switch (S11) depressed.	Defective switch (S11). Defective fuse (F1).	Replace switch. Replace fuse.
Excessive resistance J2-4 to J1-5 with CONTROL POWER switch (S11) depressed.	Defective switch (S11). Defective light bulb (L1). (Check resistance TP4 to TP5). Defective light socket (L1) (Check resistance TP4 to TP5). Defective fuse (F1).	Replace switch. Replace bulb. Replace socket. Replace fuse.
Excessive resistance J2-1 to J1-1 with switch (S1) NORMAL.	Defective switch (S1).	Replace switch.
Excessive resistance J2-2 to J1-1 with switch (S1) TEST.	Defective switch (S1). Defective resistor (R2).	Replace switch. Replace resistor.

Table 2. Fault Isolation Information (Cont)

Symptom	Probable Cause	Remedy
Excessive resistance J2-8 to J1-8 with switch (S5) NORMAL.	Defective switch (S5).	Replace switch.
Excessive resistance J2-9 to J1-8 with switch (S5) TEST.	Defective switch (S5). Defective resistor (R5).	Replace switch. Replace resistor.
Excessive resistance J2-6 to J1-6 with switch (S6) NORMAL.	Defective switch (S6).	Replace switch.
Excessive resistance J2-7 to J1-6 with switch (S6) TEST.	Defective switch (S6). Defective resistor (R6).	Replace switch. Replace resistor.
Excessive resistance J2-17 to J1-17 with switch (S4) NORMAL.	Defective switch (S4).	Replace switch.
Excessive resistance TP17 to J1-17 with switch (S4) NORMAL.	Defective switch (S4). Defective resistor (R11).	Replace switch Replace resistor.
Excessive resistance J2-19 to J1-17 with switch (S4) TEST.	Defective switch (S4). Defective resistor (R7).	Replace switch. Replace resistor.
Excessive resistance J2-18 to J1-18 with switch (S4) NORMAL.	Defective switch (S4).	Replace switch.
Excessive resistance J2-19 to J1-18 with switch (S4) TEST.	Defective switch (S4). Defective resistor (R8).	Replace switch. Replace resistor.

Table 2. Fault Isolation Information (Cont)

Symptom	Probable Cause	Remedy
Excessive resistance J2-35 to J1-35 with switch (S7) NORMAL.	Defective switch (S7).	Replace switch.
Excessive resistance J2-55 to J1-35 with switch (S7) TEST and (L2) light bulb removed.	Defective switch (S7). Defective resistor (R4).	Replace switch. Replace resistor.
Excessive resistance J2-35 to J1-55 with switch (S7) NORMAL and (L2) light bulb installed.	Defective light bulb (L2). Defective light socket (L2).	Replace bulb. Replace socket.
Excessive resistance J2-24 to J1-24 with switch (S2) NORMAL.	Defective switch (S2).	Replace switch.
Excessive resistance J2-25 to J1-24 with switch (S2) TEST.	Defective switch (S2). Defective resistor (R10).	Replace switch. Replace resistor.
Excessive resistance J2-26 to J1-26 with switch (S3) NORMAL.	Defective switch (S3).	Replace switch.
Excessive resistance J2-27 to J1-26 with switch (S3) TEST.	Defective switch (S3). Defective resistor (R9).	Replace switch. Replace resistor.
Excessive resistance J2-10 to J1-10 with switch (S10) NORMAL.	Defective switch (S10).	Replace switch.

Table 2. Fault Isolation Information (Cont)

	Symptom	Probable Cause	Remedy
	Excessive resistance J2-11 to J1-10 with switch (S10) TEST.	Defective switch (S10). Defective resistor (R12).	Replace switch. Replace resistor.
	Excessive resistance J2-15 to J1-15 with switch (S8) NORMAL.	Defective switch (S8).	Replace switch.
	Excessive resistance J2-16 to J1-15 with switch (S8) TEST.	Defective switch (S8). Defective resistor (R1).	Replace switch. Replace resistor.
	Excessive resistance J2-3 to J1-3 with switch (S9) NORMAL.	Defective switch (S9).	Replace switch.
	Excessive resistance J2-49 to J1-3 with switch (S9) TEST.	Defective switch (S9). Defective resistor (R3).	Replace switch. Replace resistor.
	Excessive resistance J2-34 to J1-34 with 3.0 volt battery disconnected.	Defective resistor (R14).	Replace resistor.
	Excessive resistance in wiring between corresponding pins and sockets of J1 and J2.	Defective wiring.	Repair or replace wiring.
R	(Code C) Excessive resistance J2-1 to J1-4 with switch (S12) in TEST position.	Defective switch (S12). Defective wiring	Replace switch. Repair or replace wiring.

R Table 2. Fault Isolation Information (Cont)

Symptom	Probable Cause	Remedy
(Codes B, C) Excessive resistance J2-13 to J1-13	Defective wiring. Defective connection at TP13.	Repair or replace wiring. Repair connection.
Excessive resistance in wiring between corresponding pins and sockets of connectors P1 and P11A or P2 and J11A.	Defective wiring. Defective connection at pin or socket.	Repair or replace wiring. Repair connection.

2. C. Perform Interstage Turbine Temperature (ITT) Component Simulation Test

(1) Test ITT simulation components Q1, CR1, R13, R14, and R15 as follows.

(a) Ensure variable resistor (R13) is set to a full counterclockwise position.

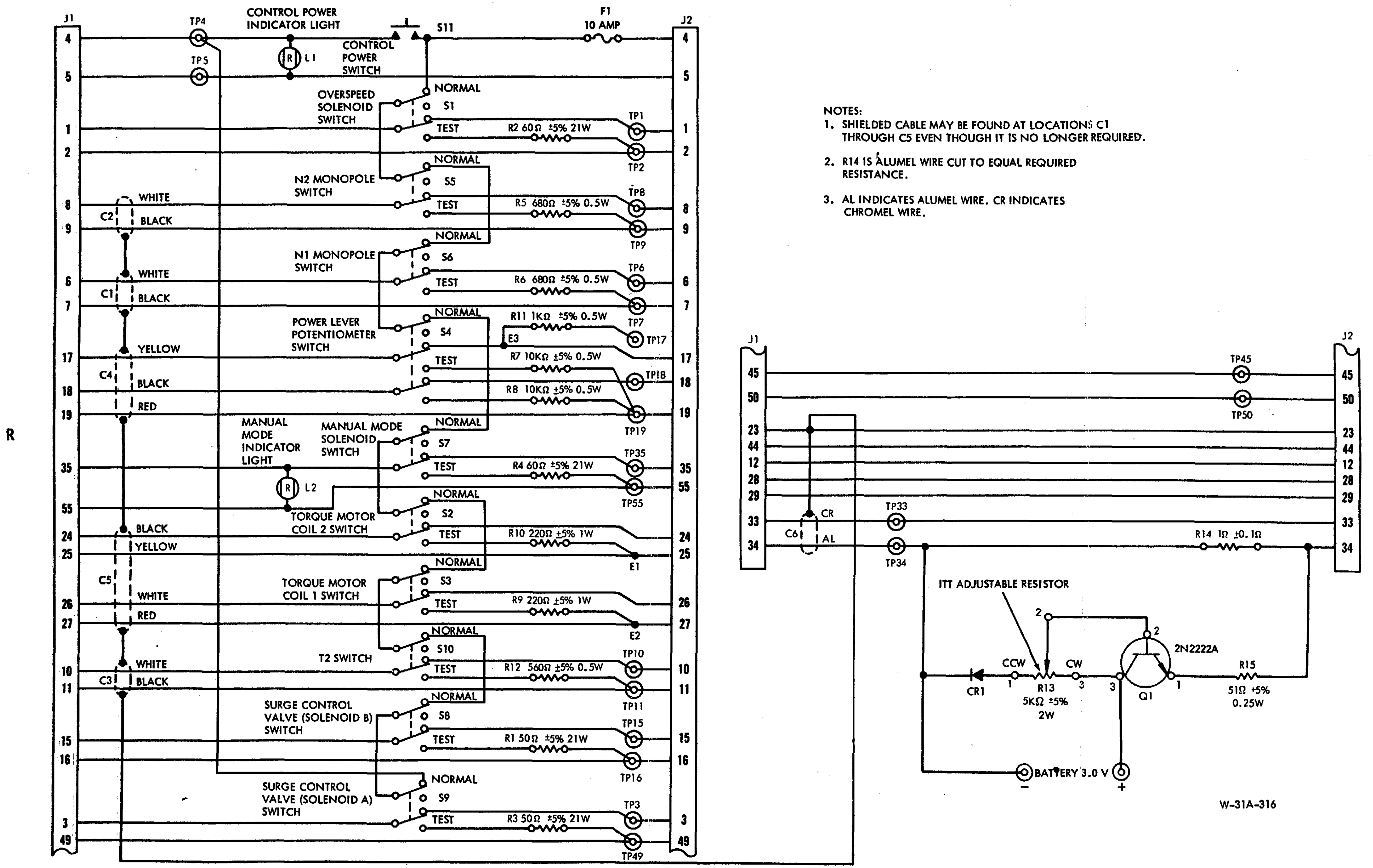
R (b) Connect external 3.0 volt battery to (+) and (-) jacks in the ITT section.

(c) Attach short jumper wire between sockets 33 and 34 of connector J2 on the tester panel.

(d) Connect millivoltmeter (Simpson Model 387 or equivalent) to plug TP33 (+) and TP34 (-).

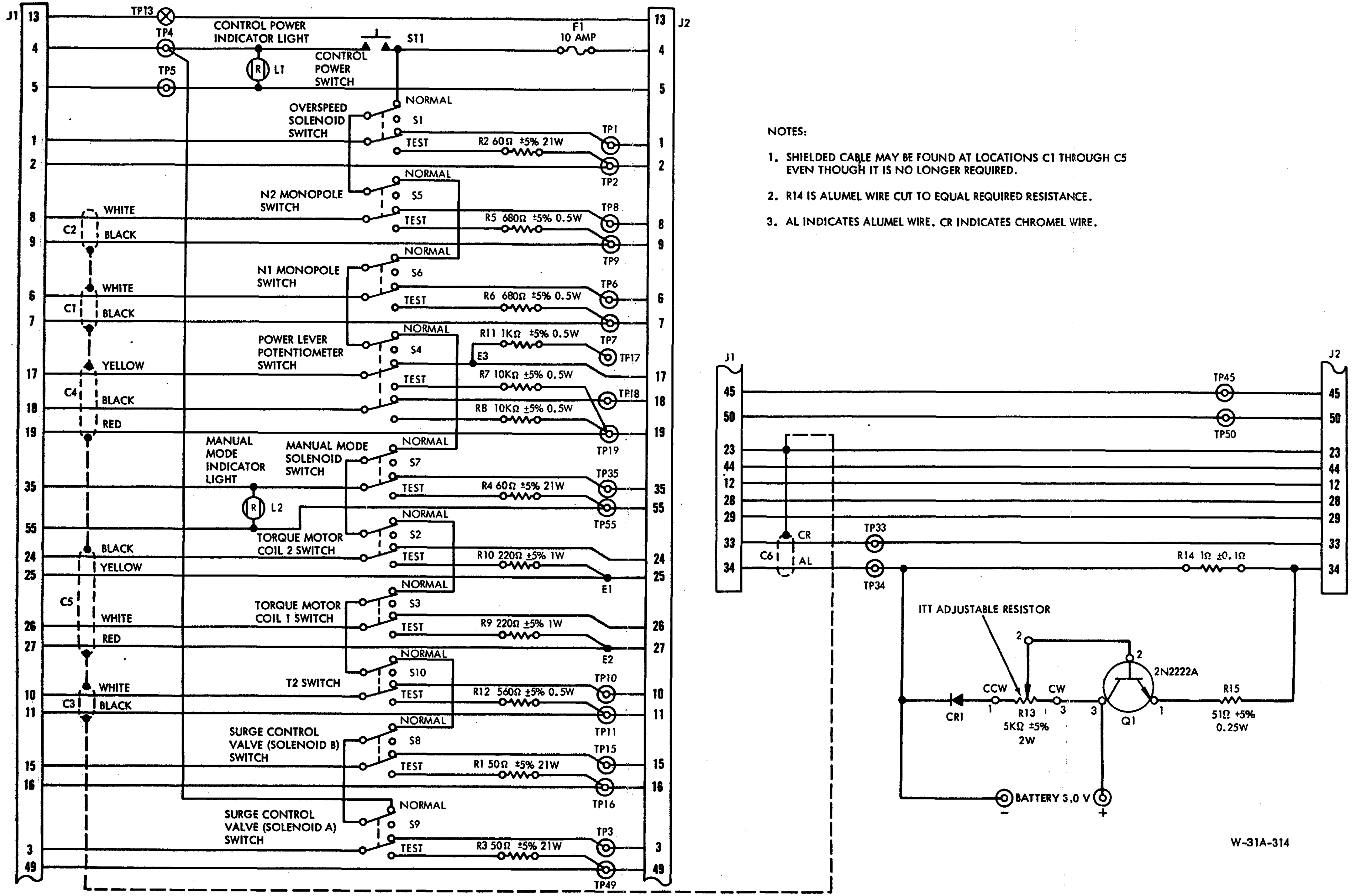
(e) Note reading on millivoltmeter scale. Turn R13 through its entire range slowly and steadily. Note deflection on the millivoltmeter scale which will vary from zero to 38 millivolts depending upon the setting of R13.

(2) Voltage reading obtained is voltage drop across the one ohm resistor in the alumel lead (R14). Each millivolt is equivalent to one milliampere of current flowing in the loop formed by the battery, transistor Q1, and resistors R14 and R15.



(Code A)
Tester Circuit Identification
Figure 1

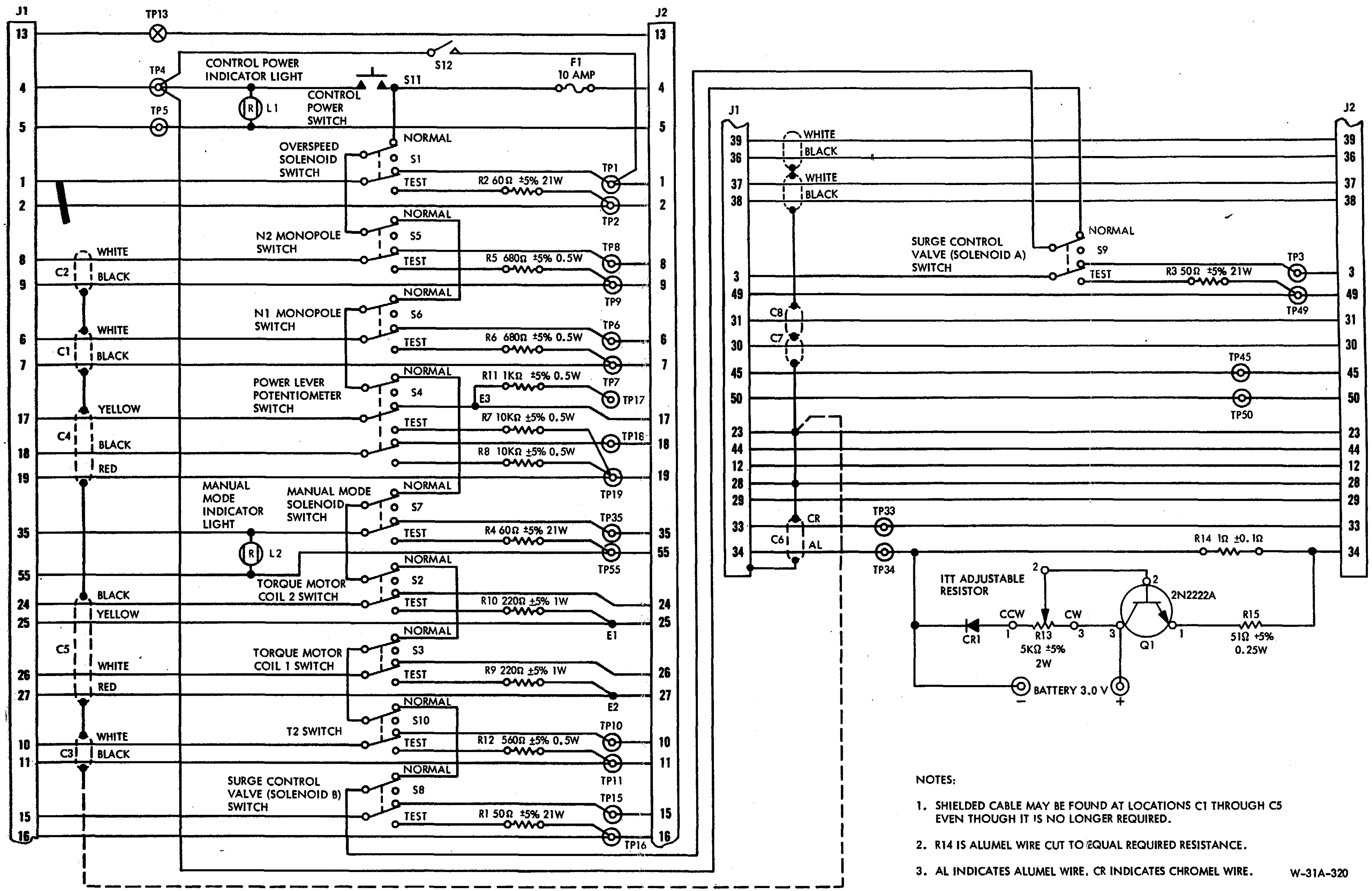
UP13342



NOTES:

1. SHIELDED CABLE MAY BE FOUND AT LOCATIONS C1 THROUGH C5 EVEN THOUGH IT IS NO LONGER REQUIRED.
2. R14 IS ALUMEL WIRE CUT TO EQUAL REQUIRED RESISTANCE.
3. AL INDICATES ALUMEL WIRE. CR INDICATES CHROMEL WIRE.

(Code B)
Tester Circuit Identification
Figure 2



NOTES:

1. SHIELDED CABLE MAY BE FOUND AT LOCATIONS C1 THROUGH C5 EVEN THOUGH IT IS NO LONGER REQUIRED.
2. R14 IS ALUMEL WIRE CUT TO EQUAL REQUIRED RESISTANCE.
3. AL INDICATES ALUMEL WIRE. CR INDICATES CHROMEL WIRE.

(Code C)
Tester Circuit Identification
Figure 3

MAINTENANCE

3. Removal/Installation

A. Remove Panel Assembly from Tester Case

R (1) Remove four screws securing panel assembly to tester case.

R (2) Lift panel assembly out of case. Electrical components mounted on
R panel assembly may be removed, replaced, or repaired using standard
R electronic shop practices. Refer to Repair, 2-4.

R B. Install Panel Assembly in Tester Case

R (1) Carefully place panel assembly into tester case. Ensure that elec-
R trical wiring or components are not caught between panel and case
R or short out against case.

R (2) Install and tighten four screws securing the panel assembly to
R case.

MAINTENANCE

4. Repair

R A. Malfunctioning Components

R Defective tester components, wiring, or connectors discovered during
R troubleshooting procedures, may be repaired or replaced in accordance
R with standard electronic shop practice.

R B. Tester Modifications

R Modifications to tester, specified by Service Bulletins, shall be made
R in accordance with the procedures outlined therein and in accordance
R with standard electronic shop practice.

R C. Tester Wiring Diagrams

R Schematic wiring diagrams for testers, and cable assemblies, are
R provided to assist in the repair or replacement of defective units.
R (See 2-4, Figures 1, 2, 3, and 4.)

R D. Replacement of Parts

R An illustrated Parts Breakdown is shown in Chapter 4 to aid in the pro-
R curement of replacement components.

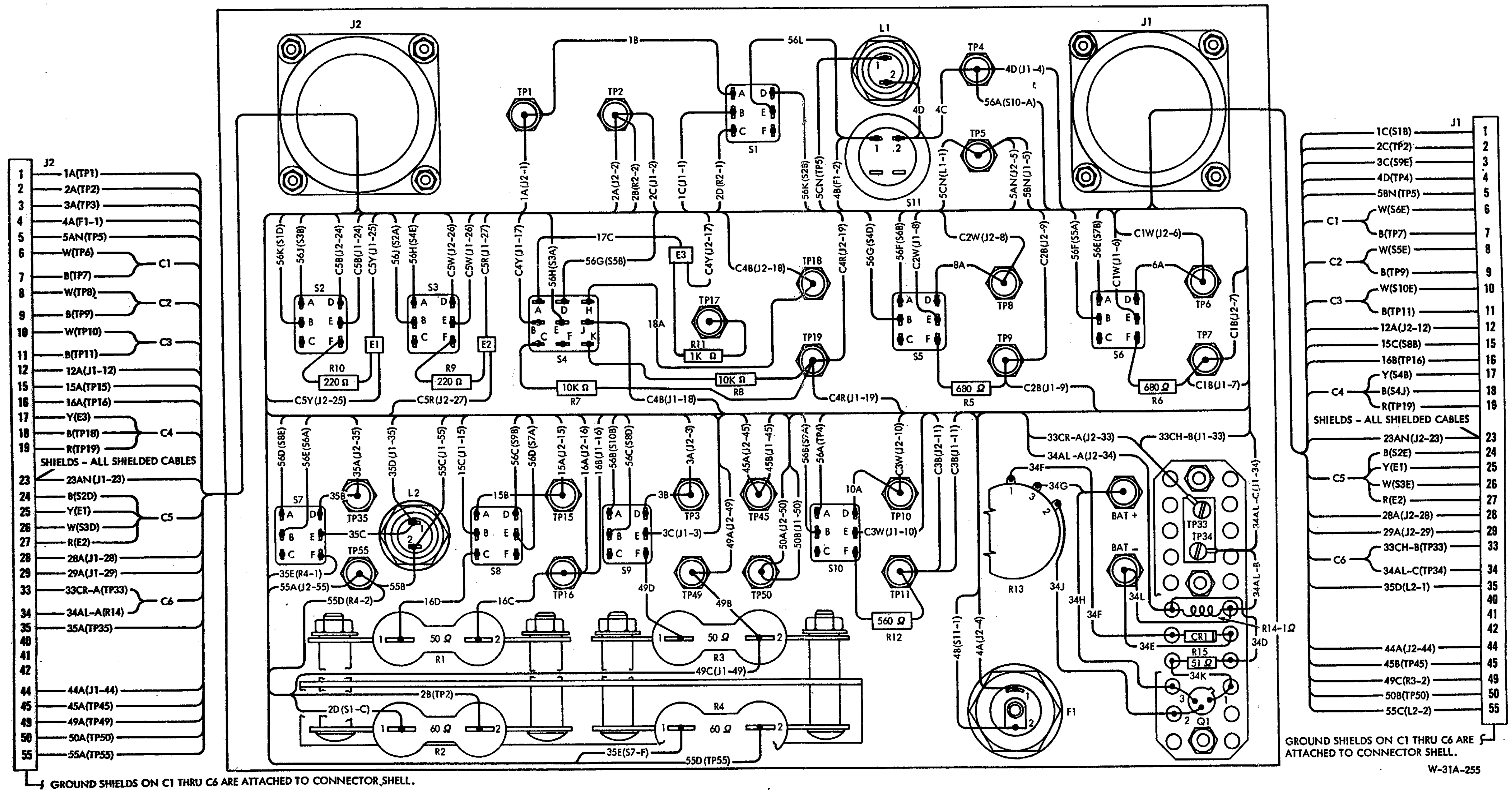
R E. Tools and Equipment

R Tools and equipment listed in 2-4, Table 1, or equivalent substitutes,
R can be used to assist in the repair of electrical circuits.

R Table 1. Equipment and Materials

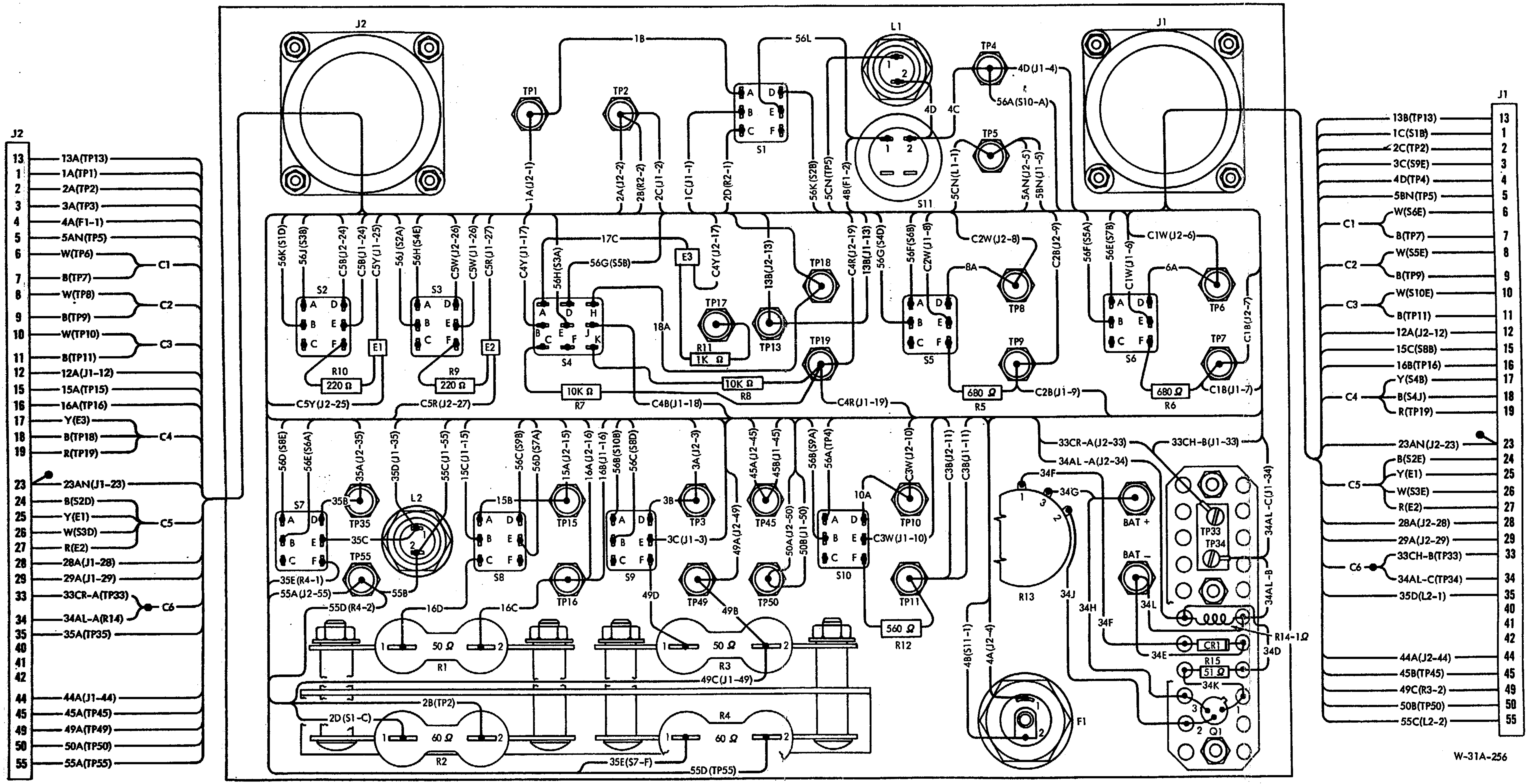
R Equipment or Materials	Manufacturer/Description
R NOTE: Equivalent substitutes may be used for listed items.	
R Millivoltmeter (Model 387)	Simpson Electric Company, 5200 W. Kinzie Street, Chicago, IL 60644 Measures voltage of ITT Signal Simulation Circuit.
R Multimeter (Type 3-3022)	Triplett Corporation, 286 Harmon Place, Bluffton, OH 45817. Measures resistance of electrical component.
R Battery (3.0 volts dc)	Commercially available. Provides power required to perform ITT component simulation test when connected to TP33 (+) and TP34 (-).

NOTE: GROUND SHIELDS ON C1, C2, C3, C4, C5 AND C6 ARE ATTACHED TO J1 AND J2 AT PIN 23.



(Code A)
Tester Wiring Diagram
Figure 1

NOTE: SHIELDED CABLE MAY BE FOUND AT LOCATIONS C1 THROUGH C5 EVEN THOUGH IT IS NO LONGER REQUIRED.



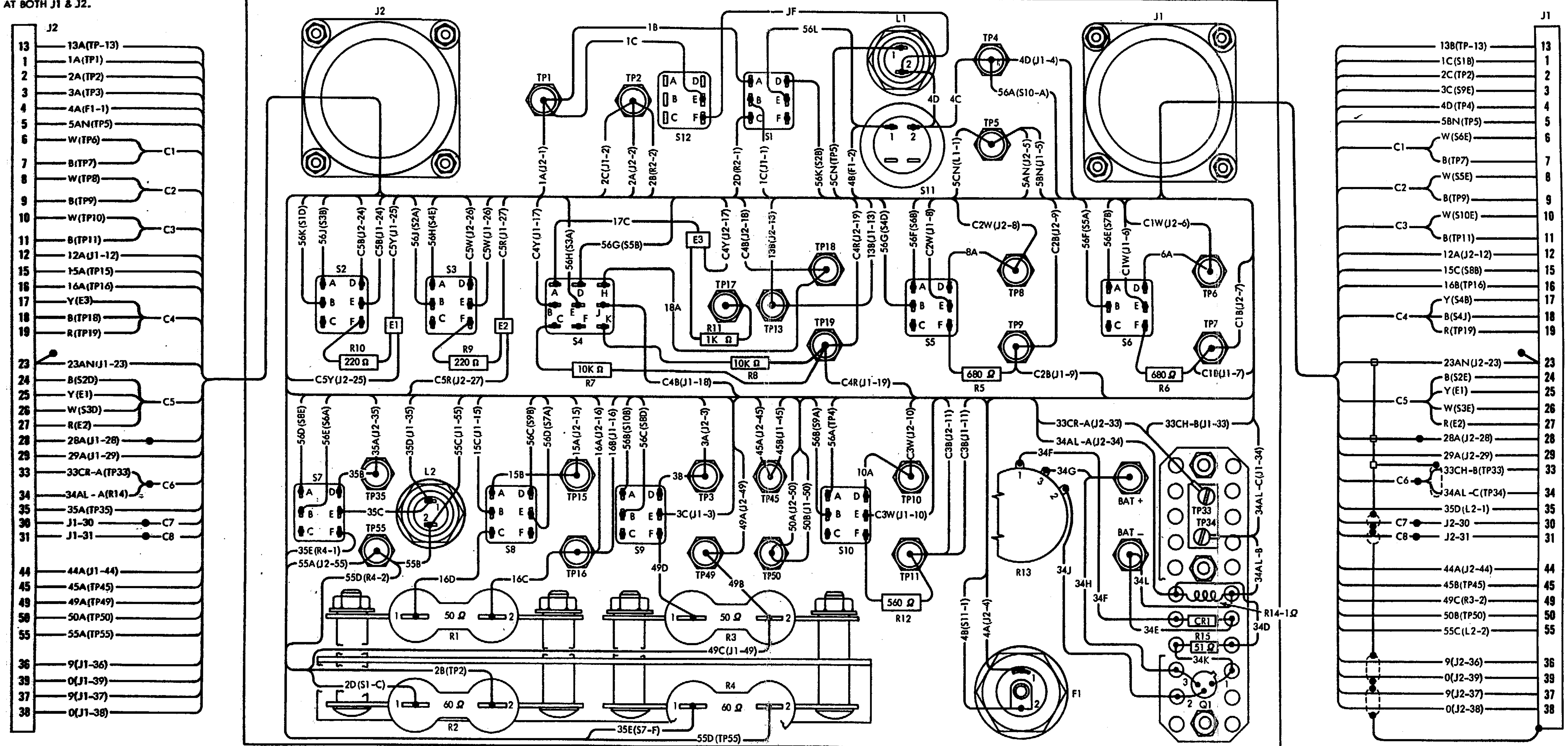
R

R
R
R

(Code B)
Tester Wiring Diagram
Figure 2

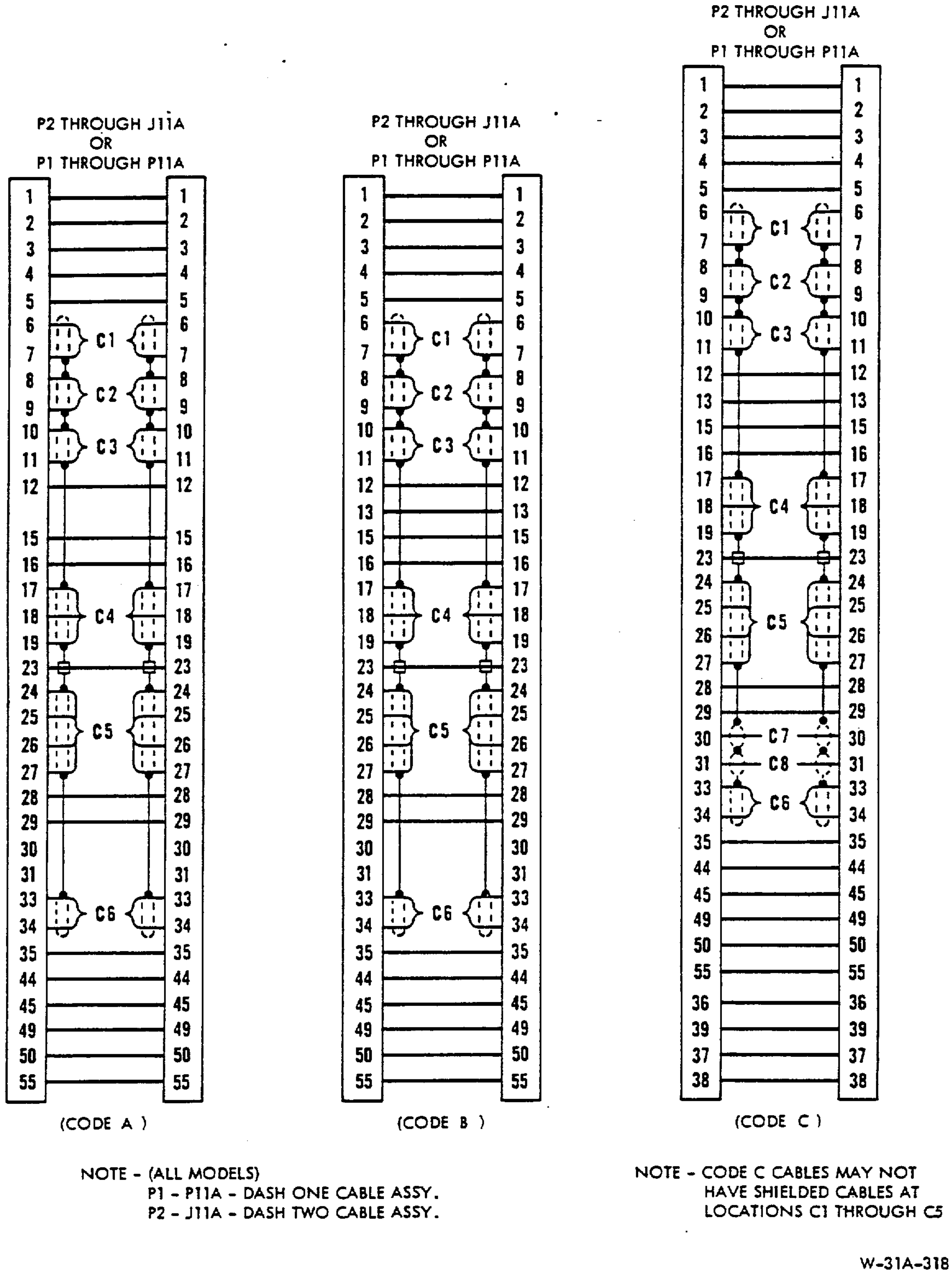
NOTE:
SHIELDED CABLE MAY BE FOUND AT LOCATIONS C1 THROUGH C5 EVEN THOUGH IT IS NO LONGER REQUIRED.

SHIELDED CONNECTIONS AT C6, C7 & C8 GROUNDED TO CONNECTOR AT PIN 23 AT BOTH J1 & J2.



W-31A-319

(Code C)
Tester Wiring Diagram
Figure 3



R

Cable Assembly Wiring Diagram
Figure 4

CHAPTER 3

TABLE OF CONTENTS

1. Overhaul

No additional instructions beyond those included in Chapter 2, MAINTENANCE, are required for the overhaul of the tester.

CHAPTER 4

TABLE OF CONTENTS

ILLUSTRATED PARTS LIST

1. General

A. Introduction

(1) Garrett part numbering system.

(a) Part numbers are assigned to each of the articles manufactured by Garrett. The part numbering consists of a straight numerical system of four through seven digits. The part numbers may be prefixed with letters and suffixed with dash numbers or letters.

(b) Prefix letters used with a part number usually indicate a type of part or method of procurement. Suffix numbers used with a part number usually indicate configuration differences or, when applied to standard part numbers, indicate design differences. Suffix letters usually indicate material, color, or finish differences.

(c) The letter "S" prefixed to a part number designates an Garrett standard part number. These standard part numbers usually are suffixed by letter and numbers.

(d) The phrase "NONPROC" followed by numbers, if any, is used in the parts list when no part number exists for an item or assembly. These entries are used solely to allow machine preparation of the parts list. The numerical portion, if any, of these entries is assigned in sequence as needed and may not appear sequentially. "NONPROC" components are not procurable and should not be ordered. When the "NONPROC" component is an assembly, the details of the assembly are procurable and may be ordered unless otherwise specified.

(e) The commercial standard numbers consist of a ten-digit number separated into two groups of three digits each and one group of four digits. A commercial standard number is a Garrett number used to identify off-the-shelf items.

(2) How to use the Illustrated Parts List.

(a) This Illustrated Parts List consists of a Numerical Index and a Detailed Parts List with exploded views. The Numerical Index relates part numbers to the figure and item number employed in the Detailed Parts List. The Detailed Parts List is keyed to the exploded views of the equipment.

1. A. (2) (b) To find the illustration of a part when the part number is known, refer to the Numerical Index and locate the part in the Part No. column. The figure and item number are found opposite the part number. Identification of the part is made by locating the item number in the appropriate figure.

(c) To identify a part when the part number is not known, turn to the Exploded View Illustrations and associated parts lists. Note the part configuration and location within the assembly. Read the part number and nomenclature off the parts list opposite the appropriate item number.

(3) Item numbers preceded by a dash (-) in the Detailed Parts List are not illustrated.

(4) Vendors' code numbers preceded by the letter "V" are used in the Nomenclature column of the Detailed Parts List to identify the
R manufacturer of a part that is purchased by Garrett. Refer to Table 1 for manufacturer's name and address.

(5) Replaced parts are identified in the Nomenclature column of the parts list by the notation REPL BY XXXXXX. The new part is identified by the notation REPL XXXXXX. A part, subassembly, assembly, or unit is replaced when the former will eventually be forced out of use by attrition. OLD and NEW are interchangeable.

(6) When a component is covered in a separate overhaul publication, the applicable publication number shall be carried in the Nomenclature column of the Detailed Parts List.

(7) Effectivity codes are used to show the difference in parts within the various configurations. Refer to Table 2.

B. Vendor Code List

Table 1. Vendor Code List

Code	Vendor
V93768	Claud S. Gordon Co., 5710 Kenosha Street, Richmond IL 60071 (R)
V99195	Thermo Electric Co, Inc, 109 Fifth Street, Saddle Brook, NJ 07662

1. C. Effectivity Code List

Table 2. Equipment Identification List
for Turbofan Engine Trouble Shooting Tester Assembly

	Code Symbol*	Part No.	Service Bulletin
R	A	289900-1	-
R	B	289900-2	289900 No. 1
R	C	289900-3	289900 No. 2

*These Alpha code symbols are used in the Effectivity Code Column of the Illustrated Parts List, and in the text, to show differences between equipment configurations.

1. D. Equipment Designator Index

Refer to Table 3 for assistance in locating items assigned equipment designators.

Table 3. Equipment Designator Index

Equipment Designator	Figure, Item	Equipment Designator	Figure, Item
CR1	4-55	S1, S3, S5, S10	3-20
E1, E2	4-95	S4	3-30
E3	4-105	S11	3-120
F1	3-15	S12	3-25
Q1	4-45	TP1, TP3, TP4, TP15	3-35
R1, R3	4-20	TP16, TP35, TP45, TP49	
R2, R4	4-25	TP+	
R5, R6	4-85	TP2, TP5, TP50, TP55	3-40
R7, R8	4-110	TP-	
R9, R10	4-100	TP6, TP7, TP8, TP9	3-45
R11	4-90	TP17, TP18, TP19	3-50
R12	4-30	TP13	3-55
R15	4-50	TP10, TP11	3-60
J1	3-75	TP33, TP34	4-65
J2	3-90		
L1, L2	3-125		

2. Numerical Index

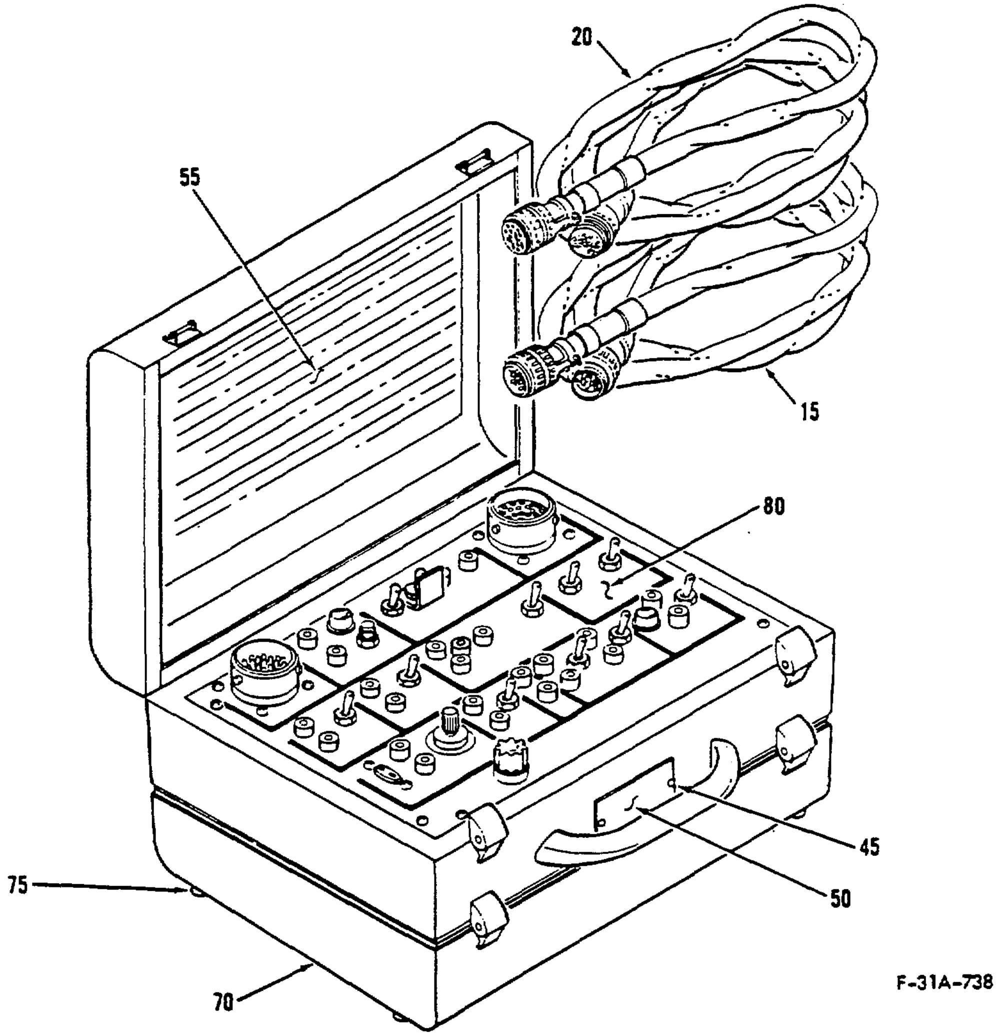
PART NO.	AIRLINE STOCK NO.	CH-SECT-UNIT FIG.	ITEM	TTL REQ
AN960-10L		4	10	4
AN960-6L		3	140	3
AN960PD4		3	110	8
K24-2SS305		4	75	AR
LH7311LC12RN2		3	125	2
MS2275-915-209		2	140	AR
MS24255-20S		4	35	2
MS24264R22T55P6		3	75	1
MS24264R22T55SN		3	90	1
MS24265R22T55PN		2	45	1
MS24266R22T55PN		2	40	1
MS24266R22T55SN		2	35	1
MS24266R22T55S6		2	30	1
MS24621-2		1	45	2
MS25036-148		2	95	2
		3	76	1
		4	80	2
MS25089-3C		3	120	1
MS25237-327		3	130	2
MS27186-1		2	70	AR
MS27291-6		2	90	2
MS3367-1-9		4	115	AR
MS3368-1-9A		2	85	2
MS3368-5-9		2	110	12
MS35207-268		4	15	4
MS35214-13		3	115	8
MS35214-29		3	150	3
MS35649-245B		3	105	8
MS35649-265B		3	135	3
MS35650-302		4	5	4
RW20V500		4	20	2
RW20V600		4	25	2
S8904B1P1		1	50	1
S8928G42E		2	135	AR
S8929G024E		2	130	AR
S8929G09E		2	115	AR
S8935G		2	145	AR
S8981E20S		2	120	AR
S8991B14L550M		2	75	AR
S9071-7-400		2	80	AR
S9352-4-0-102		2	100	12
S9510G95W		2	125	AR

PART NO.	AIRLINE STOCK NO.	CH-SECT-UNIT FIG. ITEM		TTL REQ
158-502-9101		1	75	4
224-516-9001		2	50	2
		3	95	1
224-516-9002		2	55	2
		3	100	1
224-516-9003		2	60	2
		3	80	1
224-516-9004		2	65	2
		3	85	1
23157-2KX		4	70	1
279-507-9002		3	70	1
289900-1		1	1	RF
289900-2		1	5	RF
289900-3		1	10	RF
290950-1		1	55	1
290952-1		1	15	1
		2	1	RF
290952-2		1	20	1
		2	5	RF
290954-1		4	120	1
290954-2		4	125	1
290954-3		4	130	1
292534-1		1	60	1
292535-1		1	25	1
		2	10	RF
292535-2		1	30	1
		2	15	RF
295117-1		1	35	1
		2	20	RF
295117-2		1	40	1
		2	25	RF
295146-1		1	65	1
295650-1		3	26	1
295834-1		1	65A	1
295835-1		1	35A	1
		2	26	RF
295835-2		1	40A	1
		2	27	RF
345-503-9021		3	15	1
346-507-9001		3	10	1
41102K		4	65	1
441-507-9001		3	45	4
441-507-9002		3	35	9
441-507-9003		3	40	5
441-507-9007		3	60	2

PART NO.	AIRLINE STOCK NO.	CH-SECT-UNIT FIG.	ITEM	TTL REQ
441-507-9010		3	50	3
			55	1
621-508-9045		4	85	2
621-508-9049		4	90	1
621-508-9073		4	110	2
621-521-9042		4	50	1
621-541-9071		4	100	2
621-550-9102		3	65	1
628-528-9043		4	30	1
653-520-9004		4	55	1
671-504-9007		3	145	3
672-503-9002		4	95	2
			105	1
672-503-9003		2	105	6
672-503-9004		2	111	2
684-546-9002		3	20	9
			25	1
684-546-9003		3	30	1
725-519-9002		4	40	1
734-510-9101		4	45	1
866440-1		4	60	1

3. Detailed Parts List and Exploded Views

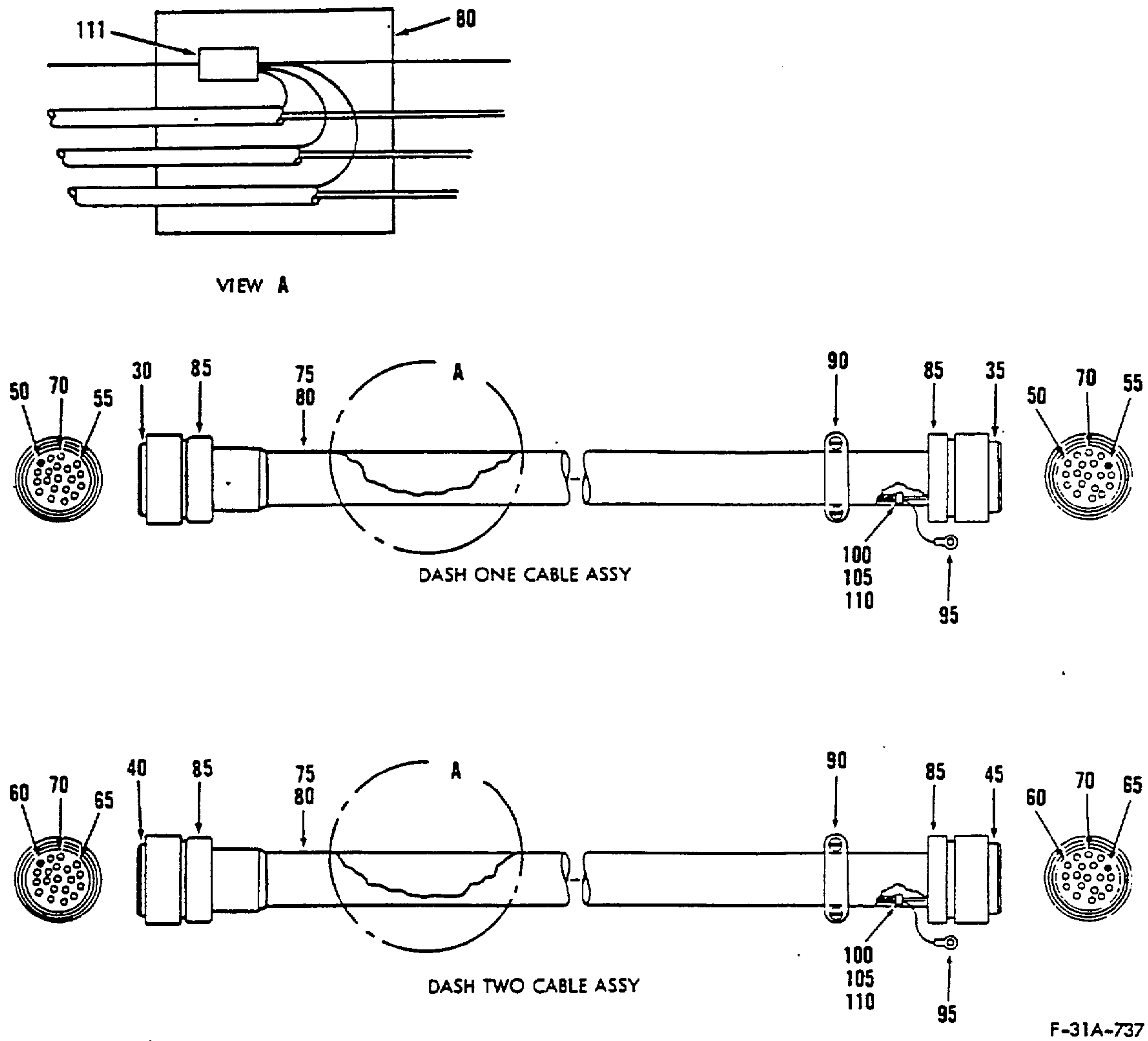
R



Turbofan Engine Trouble Shooting Tester Assembly
Figure 1

FIG. ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFFECT (USE) CODE	UNITS PER ASSY	
1						
- 1	289900-1		TESTER ASSY-TFE TROUBLE SHOOTING (DELETED BY SB 289900 NO. 1)	A	RF	
- 5	289900-2		TESTER ASSY-TFE TROUBLE SHOOTING (ADDED BY SB 289900 NO. 1) (DELETED BY SB 289900 NO.2)	B	RF	
- 10	289900-3		TESTER ASSY-TFE TROUBLE SHOOTING (ADDED BY SB 289900 NO. 2)	C	RF	
15	290952-1		.CABLE ASSY-TFE TESTER (SEE FIG. 2 FOR DETAIL)	A	1	
20	290952-2		.CABLE ASSY-TFE TESTER (SEE FIG. 2 FOR DETAIL)	A	1	
- 25	292535-1		.CABLE ASSY-TFE TESTER (SEE FIG. 2 FOR DETAIL)	B	1	
- 30	292535-2		.CABLE ASSY-TFE TESTER (SEE FIG. 2 FOR DETAIL)	B	1	
- 35	295117-1		.CABLE ASSY-TFE TESTER (SEE FIG. 2 FOR DETAIL) (REPL BY ITEM 35A)	C	1	R
- 35A	295835-1		.CABLE ASSY-TFE TESTER (REPLACES ITEM 35) (SEE FIG. 2 FOR DETAIL)	C	1	R
- 40	295117-2		.CABLE ASSY-TFE TESTER (SEE FIG. 2 FOR DETAIL) (REPL BY ITEM 40A)	C	1	R
- 40A	295835-2		.CABLE ASSY-TFE TESTER (REPLACES ITEM 40) (SEE FIG. 2 FOR DETAIL)	C	1	R
45	MS24621-2		.SCREW		2	
50	S890481P1		.NAMEPLATE-IDENTIFICATION		1	
55	290950-1		.MARKER-IDENTIFICATION	A	1	
- 60	292534-1		.MARKER-IDENTIFICATION	B	1	
- 65	295146-1		.MARKER-IDENTIFICATION (REPL BY ITEM 65A)	C	1	R
- 65A	295834-1		.MARKER IDENTIFICATION (REPLACES ITEM 65)	C	1	R
70	ORDERNHA		.CASE ASSY (NON PROCURABLE)		1	
75	158-502-9101		.BUMPER		4	
80	NONPROC03		.PANEL ASSY-TESTER (SEE FIG. 3 FOR DETAIL)		1	

— ITEM NOT ILLUSTRATED



R

Tester Cable Assemblies
Figure 2

FIG. ITEM	PART NUMBER	AIRLINE STOCK NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFFECT (USE) CODE	UNITS PER ASSY
2					
- 1	290952-1		CABLE ASSY-TFE TESTER (SEE FIG. 1 FOR NHA)	A	RF
- 5	290952-2		CABLE ASSY-TFE TESTER (SEE FIG. 1 FOR NHA)	A	RF
- 10	292535-1		CABLE ASSY-TFE TESTER (SEE FIG. 1 FOR NHA)	B	RF
- 15	292535-2		CABLE ASSY-TFE TESTER (SEE FIG. 1 FOR NHA)	B	RF
- 20	295117-1		CABLE ASSY-TFE TESTER (SEE FIG. 1 FOR NHA)	C	RF
- 25	295117-2		CABLE ASSY-TFE TESTER (SEE FIG. 1 FOR NHA)	C	RF
- 26	295835-1		CABLE ASSY-TFE TESTER (SEE FIG. 1 FOR NHA)	C	RF R
- 27	295835-2		CABLE ASSY-TFE TESTER (SEE FIG. 1 FOR NHA)	C	RF R
30	MS24266R22T55S6		.CONNECTOR-PLUG ELECTRIC (UNMODIFIED) (CMPNT OF ITEM 1,10,20 AND 26)		1
35	MS24266R22T55SN		.CONNECTOR-PLUG ELECTRIC (UNMODIFIED) (CMPNT OF ITEM 1,10,20 AND 26)		1
40	MS24266R22T55PN		.CONNECTOR-PLUG ELECTRIC (UNMODIFIED) (CMPNT OF ITEM 5,15,25 AND 27)		1
45	MS24265R22T55PN		.CONNECTOR-RECEPTACLE ELECTRIC (UNMODIFIED) (CMPNT OF ITEM 5,15,25 AND 27)		1
50	224-516-9001		.CONTACT-ELECTRICAL SOCKET-ALUMEL		2
55	224-516-9002		.CONTACT-ELECTRICAL SOCKET-CHROMEL		2
60	224-516-9003		.CONTACT-ELECTRICAL PIN-ALUMEL		2

— ITEM NOT ILLUSTRATED

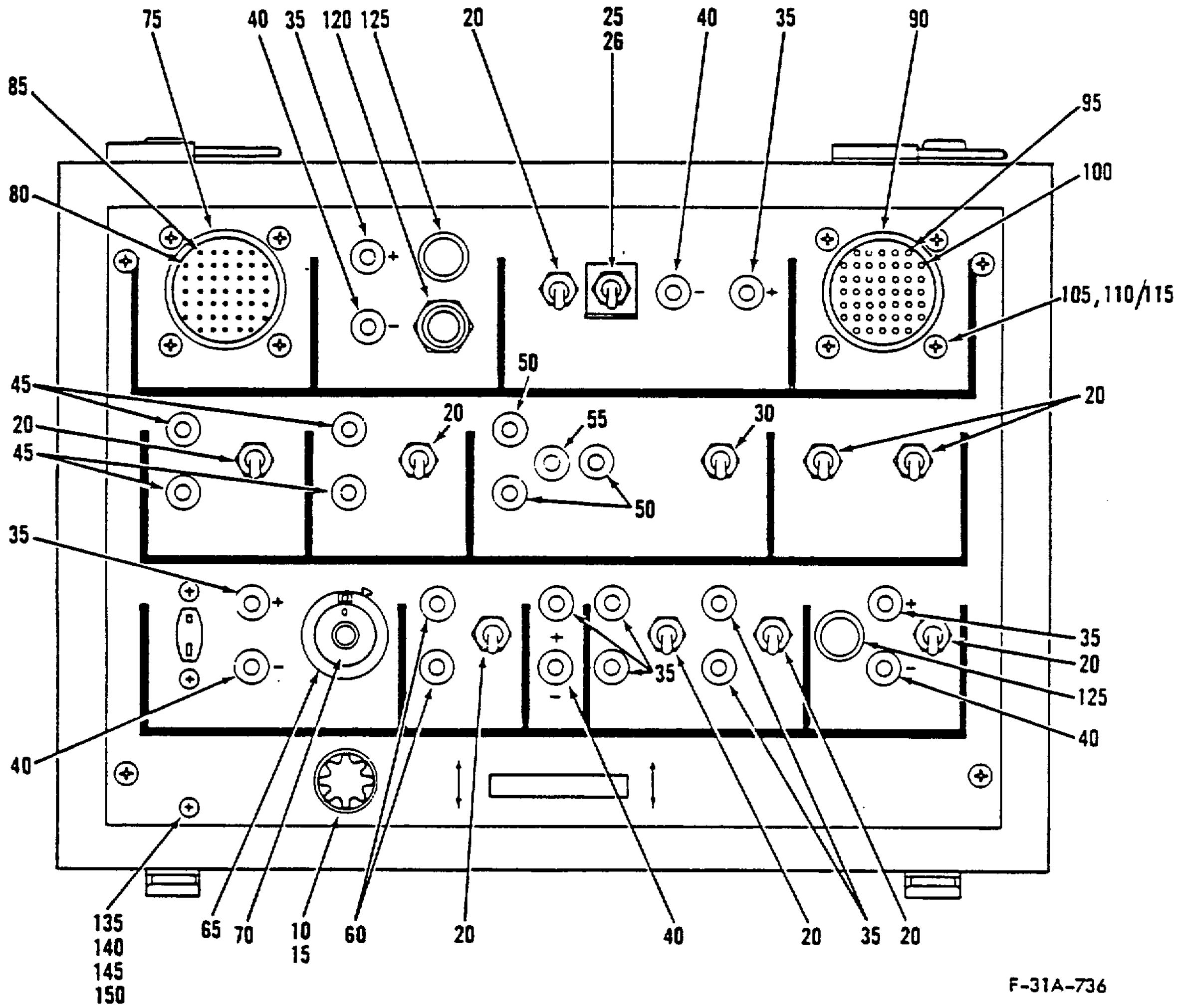
FIG. ITEM	PART NUMBER	AIRLINE STOCK NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFFECT (USE) CODE	UNITS PER ASSY
2					
65	224-516-9004		.CONTACT-ELECTRICAL PIN-CHROMEL		2
70	MS27186-1		.PLUG-END SEAL ELECTRIC CONNECTOR		AR
75	S8991B14L550M		.TUBE-RUBBER		AR
80	S9071-7-400		.INSULATION-SLEEVING		AR
85	MS3368-1-9A		.STRAP-IDENTIFICATION		2
90	MS27291-6		.SUPPORT-CABLE ELECTRIC CONNECTOR		2
95	MS25036-148		.TERMINAL-LUG		2
100	S9352-4-0-102		.INSULATION SLEEVING		12
105	672-503-9003		.SPLICE-CONDUCTOR		6
110	MS3368-5-9		.STRAP-IDENTIFICATION (CMPNT OF ITEM 1,5,10 AND 15)		12
111	672-503-9004		.SPLICE-CONDUCTOR	C	2 R
-115	S8929G09E		.CABLE-ELECTRICAL SHIELD TWO CONDUCTOR		AR
-120	S8981E20S		.CABLE-SPECIAL PURPOSE ELECTRICAL THERMOCOUPLE (CMPNT OF ITEM 1,5,10 AND 15)		AR
-125	S9510G95W		.CABLE-THERMOCOUPLE CHROMEL/ALUMEL SHIELD (CMPNT OF ITEM 20 AND 25)		AR
-130	S8929G024E		.CABLE-ELECTRICAL SHIELD THREE CONDUCTOR (CMPNT OF ITEM 1,5,10 AND 15)		AR

— ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE STOCK NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFFECT (USE) CODE	UNITS PER ASSY
2 -135	S8926G42E		•CABLE-ELECTRICAL SHIELD FOUR CONDUCTOR (CMPNT OF ITEM 1,5,10 AND 15)		AR
-140	MS2275-915-209		•WIRE (CMPNT OF ITEM 1,5,10 AND 15) (MIL-W-2275915)		AR
-145	S8935G		•WIRE-ELECTRICAL TYPE E (CMPNT OF ITEM 20,25 26 AND 27)		AR

— ITEM NOT ILLUSTRATED

R



Tester Panel Assembly
Figure 3

FIG. ITEM	PART NUMBER	AIRLINE STOCK NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFFECT (USE) CODE	UNITS PER ASSY	
3						
- 1	NONPROC03		PANEL ASSY-TESTER (SEE FIG. 1 FOR NHA)		RF	
10	346-507-9001		.FUSEHOLDER-PANEL MOUNT		1	
15	345-503-9021		.FUSE-CARTRIDGE (F1)		1	
20	684-546-9002		.SWITCH-TOGGLE (S1,S3) (S5,S10)		9	
25	684-546-9002		.SWITCH-TOGGLE (S12)	C	1	
26	295650-1		.BRACKET	C	1	R
30	684-546-9003		.SWITCH-TOGGLE (S4)		1	
35	441-507-9002		.JACK-TIP (TP1,TP3,TP4, TP15,TP16,TP35,TP45, TP49,TP+)		9	
40	441-507-9003		.JACK-TIP (TP2,TP5, TP50,TP55,TP-)		5	
45	441-507-9001		.JACK-TIP (TP6, TP7,TP8,TP9)		4	
50	441-507-9010		.JACK-TIP (TP17, TP18,TP19)		3	
55	441-507-9010		.JACK-TIP (TP13) (CMPNT OF ITEM 120 AND 125,FIGURE 4)		1	
60	441-507-9007		.JACK-TIP (TP10,TP11)		2	
65	621-550-9102		.RESISTOR-VARIABLE		1	
70	279-507-9002		.DIAL-CONTROL		1	
75	MS24264R22T55P6		.CONNECTOR-RECEPTACLE ELECTRIC (J1) (UNMODIFIED)		1	
- 76	MS25036-148		.TERMINAL LUG	C	1	R
80	224-516-9003		..CONTACT-ELECTRICAL PIN ALUMEL		1	
85	224-516-9004		..CONTACT-ELECTRICAL PIN-CHROMEL		1	
90	MS24264R22T55SN		.CONNECTOR-RECEPTACLE ELECTRIC (J2) (UNMODIFIED)		1	

— ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE STOCK NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFFECT (USE) CODE	UNITS PER ASSY
3					
95	224-516-9001		..CONTACT-ELECTRICAL SOCKET-ALUMEL		1
100	224-516-9002		..CONTACT-ELECTRICAL SOCKET-CHROMEL		1
105	MS35649-245B		.NUT		8
110	AN960PD4		.WASHER		8
115	MS35214-13		.SCREW		8
120	MS25089-3C		.SWITCH-PUSH (S11)		1
125	LH7311LC12RN2		.LIGHT-INDICATOR (L1) (L2) (MIL-L-3661/5A)		2
-130	MS25237-327		.LAMP		2
135	MS35649-265B		.NUT		3
140	AN960-6L		.WASHER		3
145	671-504-9007		.SPACER		3
150	MS35214-29		.SCREW		3
-155	NONPROC04		.PANEL ASSY-UNDERSIDE (SEE FIG. 4 FOR DETAIL)		1

— ITEM NOT ILLUSTRATED

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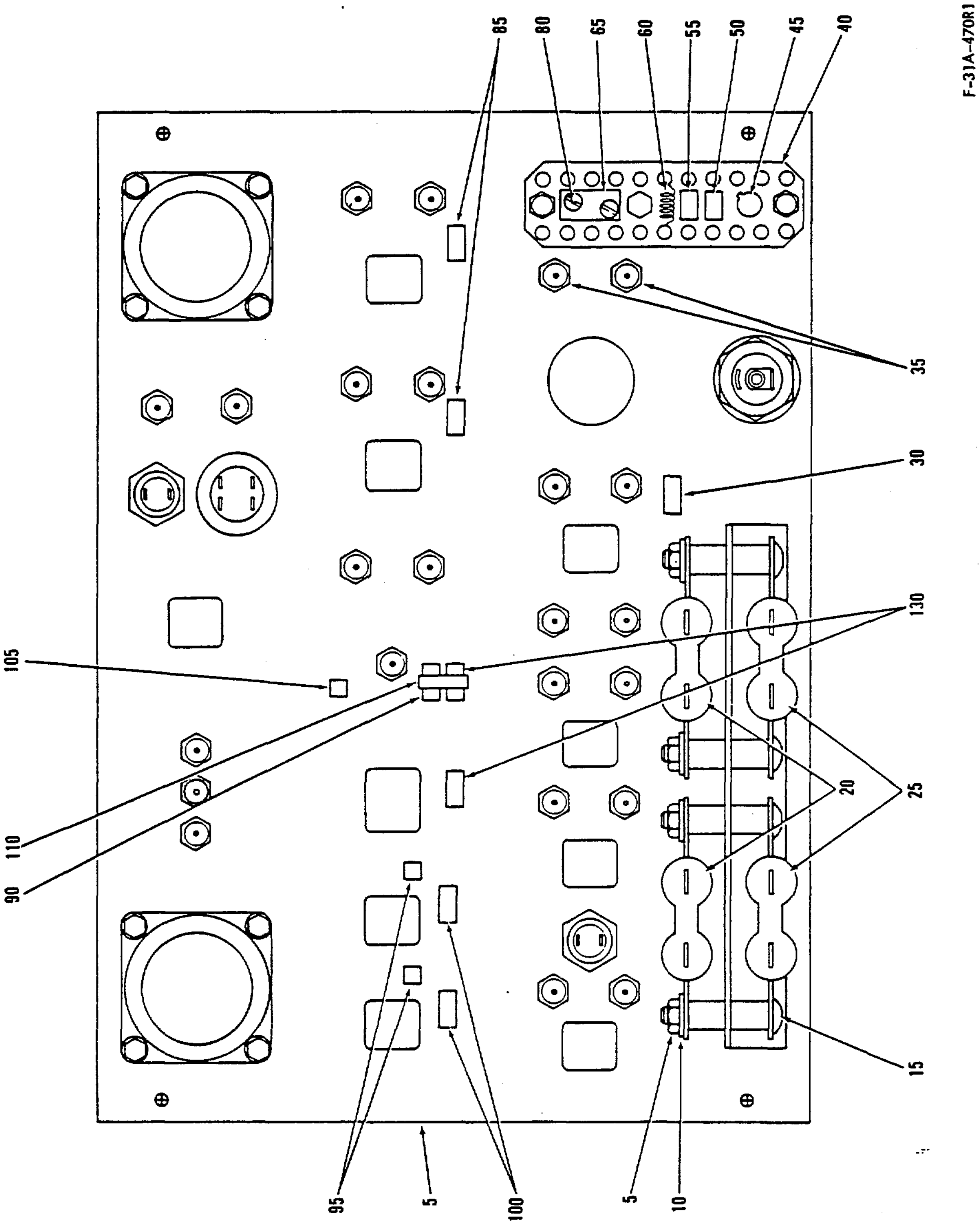

Underside Panel Assembly
Figure 4

FIG. ITEM	PART NUMBER	AIRLINE STOCK NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFFECT (USE) CODE	UNITS PER ASSY
4					
- 1	NONPROC04		PANEL ASSY-UNDERSIDE (SEE FIG. 3 FOR NHA)		RF
5	MS35650-302		.NUT		4
10	AN960-10L		.WASHER		4
15	MS35207-268		.SCREW		4
20	RW20V500		.RESISTOR (R1,R3) (MIL-R-26/2A)		2
25	RW20V600		.RESISTOR (R2,R4) (MIL-R-26/2A)		2
30	628-528-9043		.RESISTOR (R12)		1
35	MS24255-20S		.CONTACT-ELECTRICAL CONNECTOR		2
40	725-519-9002		.TERMINAL BOARD		1
45	734-510-9101		.TRANSISTOR-SWITCHING (Q1)		1
50	621-521-9042		.RESISTOR-FIXED (R15)		1
55	653-520-9004		.SEMICONDUCTOR DEVICE-DIODE (CR1)		1
60	866440-1		.RESISTOR (UNMODIFIED)		1
65	41102K		.PLUG (TP33,TP34)		1
- 70	23157-2KX		.CONNECTOR		1
- 75	K24-2SS305		.WIRE-THERMOCOUPLE		AR
80	MS25036-148		.TERMINAL-LUG		2
85	621-508-9045		.RESISTOR-FIXED (R5) (R6)		2
90	621-508-9049		.RESISTOR-FIXED (R11)		1
95	672-503-9002		.SPLICE-CONDUCTOR (E1)(E2)		2
100	621-541-9071		.RESISTOR FIXED (R9) (R10)		2
105	672-503-9002		.SPLICE-CONDUCTOR (E3)		1
110	621-508-9073		.RESISTOR-FIXED (R7) (R8)		2
115	MS3367-1-9		.STRAP-TIEDOWN		AR
-120	290954-1		.PANEL	A	1
-125	290954-2		.PANEL	B	1
-130	290954-3		.PANEL	C	1

— ITEM NOT ILLUSTRATED

CHAPTER 5

TABLE OF CONTENTS

1. Manufacturers Appendices

Not applicable for this tester assembly.